Department of Education

STEM Education
National Schools of Excellence
STEM Chemistry Syllabus
Grade 12
Term 3

DOMAIN	Phylum
SCIENCE	Chemistry
Technology	
Engineering	
Mathematics	

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SECRETARY'S MESSAGE

This Chemistry Syllabus is to be used exclusively by teachers of Chemistry to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Chemistry Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Chemistry before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principal objective of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Chemistry as a fundamental field of science.

This STEM Chemistry Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Chemistry.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Chemistry is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Chemistry to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Chemistry to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Chemistry is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Chemistry help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Chemistry is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Chemistry.

The Grade 11 Chemistry Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.
Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Chemistry through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Protein Sequencing	1. Amino Acids 2. Protein Structure and Function 3. Protein Purification 4. Protein Sequencing
HEALTH AND MEDICINE	12.1 Drug Development Process	1. Drug Targets 2. Drug Discovery 3. Drug Development and Production
ENERGY	12.1 Hydrogen – Future of Energy	1. Hydrogen as Fuel 2. Methods of Producing Hydrogen 3. Storing Hydrogen 4. Hydrogen Fuel Cell
BATTERY TECHNOLOGY	12.1 Solid-State Battery	1. Introduction to Solid-State Battery 2. Solid Electrolyte 3. Manufacturing process Solid-State Battery
ARTIFICIAL INTELLIGENCE	12.1 Artificial Molecular Machines	1. Introduction to Articifical molecular Machines 2. Catenanes and Rotaxanes 3. Other Protein based Motors under development
ROBOTICS	12.1 Robotics Meets Chemistry	1. Acidity Theories 2. Robotics 3. Programming a Robot

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;

2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;

3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and

4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Chemistry Bridging Syllabus students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Chemistry as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM DOMAIN

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science
It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology
Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering
Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics
It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM CHEMISTRY CONTENT OVERVIEW – GRADE 12, TERM 3

Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards

<table>
<tr><th colspan="4">Setting STEM Chemistry Content Standards for Progressive Learning & Assessment</th></tr>
<tr><td>Thema</td><td>Artificial Intelligence</td><td colspan="2">Module Content Standard</td></tr>
<tr><td colspan="2">Module 12.1
Artificial Molecular Machines</td><td colspan="2">12.1 To familiarize oneself with the potential of creating molecular machines.</td></tr>
<tr><td>Topics</td><td>Performance Indicators</td><td>Module Benchmark</td><td>Assessment Tasks</td></tr>
<tr><td>1. Introduction to Artificial Molecular Machines</td><td>12.1.1a: Define Artificial Molecular Machines
12.1.1b: Discuss certain chemical molecular machines that has been identified.</td><td rowspan="3">Students’ ability to master the knowledge and skills of:
• Artificial Molecular Machines
• Catenanes
• Rotaxanes</td><td rowspan="3">Assignment + Test + Quiz</td></tr>
<tr><td>2. Catenanes and Rotaxanes</td><td>12.1.2a: Explain and show with a diagram Catenanes are considered as Artificial Molecular Machines
12.1.2b: Explain and show with a diagram how Rotaxanes are considered as Artificial Molecular Machines</td></tr>
<tr><td>3. Other Protein based motors under Development</td><td>12.1.3a: Describe and explain with an example the different Artificial molecular machines that have been discovered.</td></tr>
</table>

Thema	Robotics	**Module Content Standard**	
Module 12.1 Robotics Meets Chemistry		**12.1:** Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.	
Topics	**Performance Indicators**	**Module Benchmark**	**Assessment Tasks**
1. Acidity Theories (Recall)	12.1: Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.	Students master the knowledge and skills of: • Assembling a robot • Programming a robot to solve a problem or complete a task	**Practical Assignment + Test**
2. Robotics (Basics)	12.1.1a: Explain what Bronsted-Lowry Theory, Arrhenius Theory and Lewis Acid-Base theory are and how they are related 12.1.1b: Define what pH is 12.1.1c: Explain Temperature and pH relationship		
3. Programming a robot	12.1.3a: Explain the different programming languages used in robotics		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	**Module Content Standard**	12.1	12.1
ARTIFICIAL INTELLIGENCE / 12.1 Artificial Molecular Machines	**12.1** To familiarize oneself with the potential of creating molecular machines.	✓	
ROBOTICS / 12.1 Robotics Meets Chemistry	**12.1:** Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.		✓

THEMA: ARTIFICIAL INTELLIGENCE

Introduction

In today's world, technology advances at a breakneck pace, and we are constantly exposed to new innovations. Artificial Intelligence is one of the burgeoning computer science technologies that is poised to usher in a new era in the world by creating intelligent machines. Artificial Intelligence is currently everywhere. It is now working on a wide range of subfields, from broad to specialized, such as self-driving vehicles, chess, proving theorems, music, painting, and so on.

AI is one of the most intriguing and ubiquitous areas of computer science, with a bright future ahead of it. AI could make a machine function like a person. It is made up of the terms Artificial and Intelligence, with Artificial referring to "man-made" and Intelligence referring to "thinking power," therefore AI refers to "a man-made thinking power." As a result, we may define AI as is a field of computer science that allows us to construct intelligent robots that can behave, think, and make decisions like people.

Artificial intelligence arises when a machine can learn, reason, and solve problems in the same way that humans do. You don't need to pre-program a machine to accomplish some job using Artificial Intelligence; but you may construct a computer with programmed algorithms that can work with its own intelligence, which is AI's magnificence

Artificial Intelligence is not only a branch of computer science, even though it is so broad and encompasses a wide range of other elements. To build AI, we must first understand how intelligence is made up. Intelligence is an intangible component of our brain that is made up of reasoning, learning, problem-solving vision, language comprehension, and so on.

Artificial Intelligence requires the following discipline to attain the above criteria for a machine or software:

- Mathematics
- Biology
- Psychology
- Sociology
- Computer Science
- Neurons Study
- Statistics

MODULE 12.1: ARTIFICIAL MOLECULAR MACHINES

4-5 Weeks

Machines have revolutionized civilization and improved our quality of life. What's next once we've moved from mass-producing extremely large computer machine technology to sending a man to the moon? Many chemists are now looking at the idea of constructing gadgets from molecules. Because they are man-made or "artificial," these molecules are known as Artificial Molecular Machines.

Knowing how to control motion on a miniscule scale necessitates some chemical knowledge, yet tiny machines have the potential to revolutionize everything from health to material science, where molecular processes are crucial. A machine is any device that turns energy into at least one moving part – each with its own function – and then combines these parts to produce a useful motion as an output known as "work."

Consider an old watch: all of the interconnecting cogs are arranged in such a manner that the hands on the dial move at the correct rate.

Making machines smaller has several obvious advantages, including the ability to transport them more easily and move them more precisely.

Linking Modules
Prerequisites: NIL

Broad Content Standard
12.1 To familiarize oneself with the potential of creating molecular machines.

Content Benchmark
Students' ability to master the knowledge and skills of:

- Artificial Molecular Machines
- Catenanes
- Rotaxanes

Topic 1: Introduction to Artificial Molecular Machines

Performance Indicators
Students are expected to:

12.1.1a: Define Artificial Molecular Machines

12.1.1b: Discuss certain chemical molecular machines that has been identified.

Content Expansion
Body of Knowledge for Understanding

- Brief History
- Challenges
- Chemical Bonds
- Inorganic (Chemical) Molecular Machines

Set of Proficiency Skills

- Communication Skills
- Problem-Solving Skills
- Domain Knowledge
- Time Management Skills
- Teamwork Skills
- Thirst for Learning

Traits of Attitudes and Values

- Be comfortable with failure and ability to learn from failures
- Curiosity
- Good data intuition

Topic 2: Catenanes and Rotaxanes

Performance Indicators

Students are expected to:

12.1.2a: Explain and show with a diagram Catenanes are considered as Artificial Molecular Machines

12.1.2b: Explain and show with a diagram how Rotaxanes are considered as Artificial Molecular Machines

Content Expansion

Body of Knowledge for Understanding

- Catenanes
- Rotaxanes

Set of Proficiency Skills

- Communication Skills
- Problem-Solving Skills
- Domain Knowledge
- Time Management Skills
- Teamwork Skills
- Thirst for Learning
- Software Engineering Skills
- Data Science Skills
- Machine Learning Skills

Traits of Attitudes and Values

- Be comfortable with failure and ability to learn from failures
- Curiosity
- Good data intuition

Topic 3: Other Protein based Motors Under Development

Performance Indicators

Students are expected to:

12.1.3a: Describe and explain with an example the different Artificial molecular machines that have been discovered.

Content Expansion

Body of Knowledge for Understanding

- Artificial
    - Molecular Motors
    - Molecular Propeller
    - Molecular Switch
    - Molecular Shuttle
    - Nanocar
    - Molecular Balance
    - Molecular Tweezers
    - Molecular Sensor
    - Molecular Logic Gate
    - Molecular Assembler
    - Molecular Hinge
- Biological
- Research

Set of Proficiency Skills

- Communication Skills
- Problem-Solving Skills
- Domain Knowledge
- Time Management Skills
- Teamwork Skills

Traits of Attitudes and Value

- Curiosity
- Good data intuition

THEMA: ROBOTICS

Introduction

Robotics is a multidisciplinary discipline combining computer science and engineering. It is the study of the design, building, operation, and use of robots. Robotics aims to create devices that can aid and benefit people.

Mechanical engineering, electrical engineering, information engineering, mechatronics, electronics, bioengineering, computer engineering, control engineering, software engineering, and mathematics are just a few of the areas that robotics encompasses.

Robotics is the study of creating machines that can perform the function of people and mimic their behaviors. Robots can be employed in a variety of settings and for a variety of reasons, but many are now utilized in hazardous areas (such as radioactive material inspection, bomb detection and deactivation), manufacturing operations, or in conditions where humans are unable to live (e.g. in space, underwater, in high heat, and clean up and containment of hazardous materials and radiation).

MODULE 12.1: ROBOTICS MEET CHEMISTRY

1 Week

This module is more project-based and is designed to simply have students design a Chembot – a robot that does simple chemistry experiments. The aim is to have students see the link between branches of science – in this case, chemistry with robotics, mechanical engineering, electrical engineering, and programming. The challenge presented in this module is for students to build a robot to answer the question: “When temperature changes, does it affect the pH of a liquid or not?”

Linking Modules
Prerequisites: NIL

Broad Content Standards
12.1: Design and build a Chembot that tells whether or not the pH of a liquid as changed with a change in temperature.

Content Benchmarking
Students master the knowledge and skills of:

- Assembling a robot
- Programming a robot to solve a problem or complete a task

Topic 1: Acidity Theories (Recall)

Performance Indicators

Students are expected to:

12.1.1a: Explain what Bronsted-Lowry Theory, Arrhenius Theory and Lewis Acid-Base theory are and how they are related

12.1.1b: Define what pH is

12.1.1c: Explain Temperature and pH relationship

Content Expansion

Body of Knowledge for Understanding

- Overview
- Acids
- Factors affecting acidic strength
    - Polarity of the molecule and strength of a H-A bond
- Bases
- Arrhenius acid-base theory (water-ion system)
    - Some Arrhenius acids
    - Some Arrhenius Bases
    - Neutralization Reaction
- Utility of Arrhenius concept
- Hydrogen ion (H+) or Hydronium Ion (H_3O^+)
- Concept of pH
- Amphoteric Nature of Water
- Advantages of Arrhenius Theory
- Limitations of Arrhenius Acid-Base Theory
- Bronsted-Lowry Theory
    - Conjugated base pairs
    - Examples of Bronsted-Lowry acids and bases
    - Advantages of Bronsted-Lowry Theory
    - Disadvantages of Bronsted-Lowry Theory
- Relationship between Arrhenius Theory and Bronsted-Lowry Theory
- Lewis Acid-Base Theory
    - Lewis acid
        - Characteristics
    - Lewis base
        - Characteristics
    - Examples
    - Neutralization Reaction
    - Limitation
- Relationship between Lewis Acid-Base Theory and Arrhenius Theory

Set of Proficiency Skills

- Able to build Robots with a function

Traits of Attitudes and Values

- Thinking and working collaboratively

Topic 2: Robotics (Basic)

Performance Indicators
Students are expected to:
12.1.2a: Explain the components of robots

Content Expansion
Body of Knowledge for Understanding

- Robotics
- Components
- Robot Locomotion
    - Legged
    - Wheeled
- Types of Robots
    - Mobile
        - Rolling
        - Walking
    - Industrial
    - Autonomous
    - Remote Controlled (RC) Robots
- Types of Sensors
    - Light
        - Photo resister
        - Photovoltaic Cells
    - Proximity
        - Infrared Transceivers
        - Ultrasonic Sensor
    - Sound
    - Temperature
    - Acceleration

Set of Proficiency Skills
- Able to build Robots with a function

Traits of Attitudes and Values
- Thinking and working collaboratively

Topic 3: Programming a Robot

Performance Indicators
Students are expected to:
12.1.3a: Explain the different programming languages used in robotics

Content Expansion
Body of Knowledge for Understanding
- What is Robotics
- Fiver Major fields of Robotics
- Robotic Programming
- Software
    - Robots Operating System (ROS)
    - Robot Control Software
- Programming Language
    - Pascal
    - Scratch
    - Industrial Robot Language
    - LISP and Prolog
    - Hardware Description Language
    - MATLAB
    - C#/.NET
    - JAVA
    - Python
    - C/C++

Set of Proficiency Skills
- Able to build Robots with a function

Traits of Attitudes and Values
- Thinking and working collaboratively

REFERENCES

- Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de)
- Robot Programming: A Guide to Programming Your Own Robots (edgy.app)
- Nanorobotics (fractal.org)
- Making and Operating Molecular Machines: A Multidisciplinary Challenge - Baroncini - 2018 - ChemistryOpen - Wiley Online Library
- Driving Smart Molecular Systems by Artificial Molecular Machines (wiley.com)
- Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de)
- https://www.scienceinschool.org/article/2016/chembot-chemistry-robots/
- Robot Programming: A Guide to Programming Your Own Robots (edgy.app)
- Robotics Designing of PCB - javatpoint
- https://blog.finology.in/Legal-news/Ways-to-Commercialize-IPR
- https://www.upcounsel.com/why-do-patents-expire
- Ihttps://courses.lumenlearning.com/wmopen-introductiontobusiness/chapter/intellectual-property-2/
- https://www.forbes.com/sites/arielcohen/2021/02/11/what-batteries-will-power-the-future/?sh=5ce1b64841c0
- https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fcourses.mak.ac.ug%2Fsites%2Fdefault%2Ffiles%2Fdownloads%2FIndustrial%2520Biochemistry.doc&wdOrigin=BROWSELINK
- https://www.biochemistry.org/education/careers/becoming-a-bioscientist/what-is-biochemistry/
- https://chemintelligence.com/ai-for-chemistry#:~:text=Artificial%20Intelligence%20%28AI%29%20is%20being%20used%20more%20and,and%20the%20time%20to%20market%20for%20new%20drugs.
- https://www.javatpoint.com/facebook-data-mining